DAA ASSIGNMENT | BATCH 10

FIELD SERVICE: Assigning Technicians to Tickets



A cost-minimization study using Branch and Bound and Greedy approach

BRANCH & BOUND

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Overview of the Field Service Problem

Field service teams receive service tickets that must be handled by available technicians.

TECHNICIANS

T1 • T2 • T3 • T4

People who perform the service work.

SERVICE TICKETS

K1 • K2 • K3 • K4

Jobs that need to be completed.

ASSIGNMENT COST

Minimum total

Each matrix value represents the cost of one technician-ticket assignment.

The task is to find a one-to-one assignment with the lowest possible total cost.

Objective and Assignment Requirements

MAIN OBJECTIVE

Minimize the total cost of assigning technicians to service tickets.

Total Cost = Sum of the selected assignment costs

REQUIREMENTS

- Each technician must receive exactly one ticket.
- Each ticket must be assigned to exactly one technician.
- A ticket cannot be assigned to two technicians.
- The final solution must satisfy any additional restrictions.

Batch 10 Assignment Cost Matrix

Rows represent technicians; columns represent service tickets. A smaller value means a lower assignment cost.

Technician	K1	K2	K3	K4
T1	11	**4**	9	7
T2	**8**	6	5	10
T3	7	10	**3**	9
T4	9	8	11	**6**

The highlighted cells indicate the assignments selected in the optimal solution.

Methods Used in This Assignment

BRANCH AND BOUND

A systematic optimization technique.

- Builds partial assignment solutions.
- Calculates a lower bound for a partial solution.
- Prunes a branch when it cannot improve the best known result.
- Used to obtain the minimum-cost assignment.

GREEDY APPROACH

A step-by-step local choice method.

- Selects the lowest currently available cost.
- Makes one choice at a time.
- Does not reconsider earlier choices.
- Can be quick, but may not give the global minimum.

Optimal Assignment: Total Cost = 21

The assignment reported in the source presentation is:

T1 → K2

Assignment cost: 4

T2 → K1

Assignment cost: 8

T3 → K3

Assignment cost: 3

T4 → K4

Assignment cost: 6

4 + 8 + 3 + 6 = 21 TOTAL MINIMUM COST

Branch and Bound uses pruning to avoid exploring branches that cannot beat the best solution.

How Branch and Bound Searches

1	**Branch**	Choose a possible technician-ticket assignment.
2	**Calculate**	Estimate the minimum possible completion cost using a lower bound.
3	**Compare**	Compare the bound with the best complete solution found so far.
4	**Prune or explore**	Discard an unpromising branch; continue exploring a promising branch.

The aim is to reduce unnecessary search while preserving the best feasible solution.

Greedy Approach: Step-by-Step Execution

Step 1

T1 picks K2

Cost = 4

Free tickets: K1, K3, K4

Step 2

T2 picks K3

Cost = 5

Free tickets: K1, K4

Step 3

T3 picks K1

Cost = 7

Free ticket: K4

Step 4

T4 is forced to K4

Cost = 6

No tickets remain

Greedy total = 4 + 5 + 7 + 6 = 22

Branch and Bound vs. Greedy

Method	Assignment sequence	Total cost
Branch and Bound	T1→K2, T2→K1, T3→K3, T4→K4	21
Greedy	T1→K2, T2→K3, T3→K1, T4→K4	22

Branch and Bound

Produces the lower cost: 21

Greedy

Produces cost: 22 (one unit higher)

Greedy makes locally good choices, but these choices can restrict later assignments.

Lower Bounding Rules

A lower bound estimates the least cost that may be needed to complete a partial assignment.

EXAMPLE: T1 → K2

Current assignment cost = 4

- Remaining row minimum for T2 = 5
- Remaining row minimum for T3 = 3
- Remaining row minimum for T4 = 6

LOWER BOUND CALCULATION

4 + 5 + 3 + 6

= 18

This is a theoretical lower bound for that partial assignment, not the final assignment cost.

Two Lower Bounding Rules

RULE 1

Row Minimums

- Find the smallest available cost in each unassigned row.
- Add these minimum values to the current partial cost.
- It is fast to calculate, but may give a looser bound.

RULE 2

Matrix Reduction

- Subtract row and column minimum values.
- Use the reduced matrix to estimate the remaining cost.
- A tighter bound can help prune more branches.

Both rules help the algorithm decide which branches need further exploration.

Handling Additional Rules

The assignment may need to satisfy restrictions in addition to minimizing cost.

CONSTRAINT 1

T3 is not certified for K1.

CONSTRAINT 2

T1 and T2 must not both be assigned within the K2/K3 block.

HOW TO HANDLE THEM

- Set Cost[T3][K1] = ∞ to mark the forbidden assignment.
- Add a condition to check the K2/K3 group restriction.
- Prune any partial assignment that violates a constraint.

The source PDF does not calculate a new optimum after applying these additional constraints.

Results and Key Takeaways

FINAL COMPARISON

Branch and Bound: 21

Greedy: 22

- The assignment problem aims to minimize the total technician-ticket cost.
- Branch and Bound uses lower bounds and pruning to find the best assignment.
- Greedy is simple, but local choices may lead to a higher total cost.
- Additional real-world restrictions must be checked during assignment.

THANK YOU

Contribution Table

Each member contributed to a specific part of the DAA assignment and presentation.

Roll Number	Name	Contribution
24I362	**Sriram A.**	Repository setup, Branch and Bound (main solution) implementation, optimal assignment and pruning, pull request submission, and video presentation.
24I344	**Purushothaman S.**	Greedy Approach implementation, timing analysis script, plot generation, and video presentation.
24I340	**Praveen Kumaran**	Report and RUN.md documentation, presentation slide creation, PPT design, and video presentation.